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Paleomagnetism of the Lower Permian redbeds of the Abadla Basin (Algeria)

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Abstract

One of the reference Permian paleomagnetic poles for the African Apparent Polar Wander Path (APWP) was determined by (Morel et al., 1981, Earth Planet. Sci. Lett. 55, 65–74) in the upper unit of the 2000 m thick Abadla redbeds formation (31.2ºN, 2.7ºW). Unfortunately, this upper unit remains undated, although its age was presumed to be Autunian, whereas microflora fossils characteristic of the Autunian were discovered in the lower unit of the same formation. The paleomagnetic data ($N = 11$, $D = 130.1^{o}$, $I = 13.0^{o}$, $k = 138$, $\alpha_{95} = 3.6^{o}$ and pole: 29.1ºS, 57.8ºE, $A_{95} = 2.0^{o}$) obtained from the better dated Autunian levels must take the place of the previous Abadla pole for the Autunian part of the African APWP. This new pole is very close to the Morel et al.'s pole determined from the unfossiliferous upper unit of the Abadla formation, giving an Autunian age for this upper unit. Consequently, the subsidence of the Abadla basin and the deposition of the whole succession occurred during a period not longer than 20–25 Myr. © 1998 Elsevier Science B.V. All rights reserved.

Keywords: Africa; paleomagnetism; Autunian; subsidence; apparent polar wander path

1. Introduction

The study of the relative motion of Gondwana and Laurentia during the late Paleozoic times (Carboniferous to Permian) remains an issue of some controversy. Analysis of the geological distribution of paleoclimatic indicators and studies of regional geology and tectonics help to constrain the relative positions of these super-continents. However, only paleomagnetism can quantify the latitudinal component of motion. Unfortunately, the Apparent Polar Wander Path (APWP) of Gondwana is for the most part poorly defined and has no clear trend. In fact, the data are sparse and often affected by large uncertainties. This is mainly due to the lack of formations suitable for paleomagnetism on the cratonic Gondwana, to a poor or insufficient age control in some cases (Salmon et al., 1986) and to the importance of widespread remagnetizations.

Thus, one of the reference Permian paleomagnetic poles has been determined by Morel et al. (1981) in the upper unit of the thick (almost 2000 m) Abadla redbeds formation whose age is poorly constrained. The aim of the present study is to obtain a paleomagnetic pole for the lower unit which is now precisely dated. The comparison of this new pole with that of

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Morel et al. helps to constrain the age of the upper unit of the formation.

2. Geological setting and sampling

The Algerian Abadla basin (31.2ºN, 2.7ºW) is located on the northwestern edge of the Saharan craton, south of Bechar (Fig. 1a). The redbeds formation lies conformably on Upper Carboniferous (Westphalian) rocks containing coal. It is locally overlain also by volcanic flows or sills of the Koudiat Zerigat (Fabre, 1976) attributed to the Late Triassic or Liassic. The paleomagnetic pole of these volcanics (Kies, 1994) is poorly defined because of the probable superimposition of different components of magnetization that have not been properly isolated. It falls close to the Late Permian–Triassic section of the African APWP. The Abadla redbeds formation is capped by the Neogene sedimentary cover of the Hamada formation. The stratigraphical study of the Abadla redbeds was made by Deleau (1951) who subdivided the formation into two units, as illustrated by a N–S cross-section (Fig. 1b).

The lower unit is about 1000 m thick and is composed of red or violet sandstones, red and green clays and some interbedded limestone horizons. Deleau (1951) proposed a Stephanian age for this unit on the basis of a fossil fauna. Doubinger and Fabre (1983) collected 23 samples for a palynological study and pointed out the presence of a Stephanian and Autunian microflora respectively at levels C and E to H (Fig. 1b). The sedimentological data show that this lower unit is a rhythmic succession that deposited in lacustrine and fluviatile conditions (Fabre, 1988). Except for site 7, all the paleomagnetic samples were collected in this well dated unit (Fig. 1a,b). Sites 1 to 6 (28 sandstone and 5 marl red hand samples) and 12 to 13 (20 sandstone hand-samples) are located at Garet El Hamra, while sites 8 to 11 (29 sandstone red and green or gray hand-samples) on the North side of Hadeb Chouahed. Marl hand-samples (5) of site 4 were too soft to be drilled and were discarded.

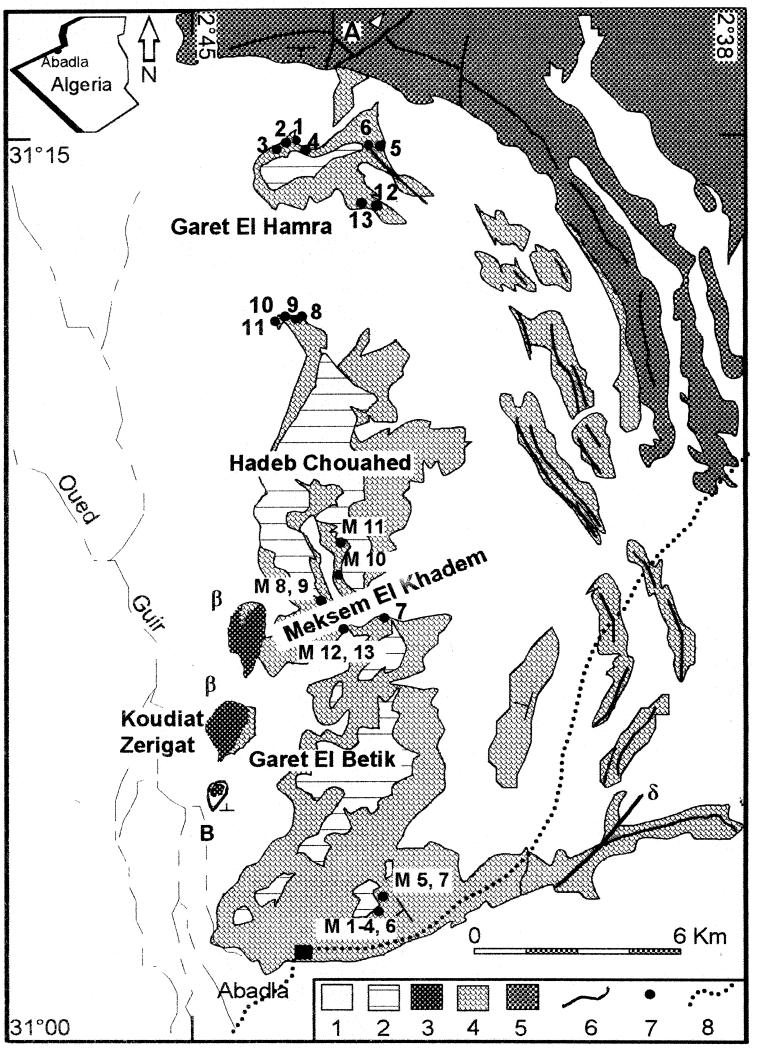
The upper unit has nearly the same thickness as the lower one and lies on it without apparent discordance (Fabre, 1988). This unit is a rhythmic succession of red clays and sandstones and has not yielded any fossil. Thirteen sites were previously studied for paleomagnetism by Morel et al. (1981). One of our sites (site 7 consisting of 9 red sandstone hand samples, Meksem El Khadem) was also taken in this unit.

The two units have almost uniform weak southward dip. The tilting occurred after the end of the deposition of the upper unit. The overlying Koudiat Zerigat volcanics are horizontal, but no contact with the underlying formation has been observed between these lavas and the redbeds (Kies, 1994).

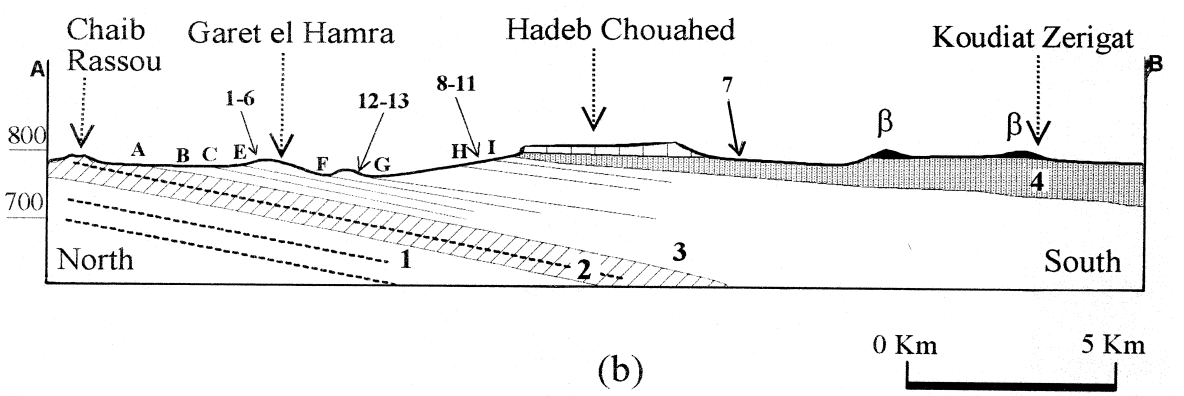
3. Laboratory and analysis procedures

One or two cores were drilled perpendicularly to the bedding from 86 large hand samples. At least two specimens were cut from each of the 167 obtained oriented cores. One specimen, from different levels of the hand sample, has been selected in each core. Before any magnetic analysis, all the specimens had been left in a zero-magnetic field shielded space for more than two weeks in order to reduce the viscous remanent magnetization (VRM) acquired in-situ and after sampling. The intensity and the direction of the natural remanent magnetization (NRM) were measured on a JR-4 spinner magnetometer (Geofyzika, Brno). Alternating magnetic field (AF) and thermal demagnetization techniques were performed by means of respectively an AF demagnetizer which automatically cancels the parasitic anhysteretic remanence (Le Goff, 1985), and a magnetically shielded furnace. Both instruments were constructed at the Saint Maur Laboratory. The results of a pilot study showed that the AF procedure was not efficient, either for a complete demagnetization or for separating

Fig. 1. (a) Geological map of Abadla basin (modified after Morel et al., 1981). *1* = Quaternary, *2* = Hamada (Neogene), *3* = Doleritic basalts (flows β and dykes δ), *4* = Abadla formation, *5* = Westphalian, *6* = faults, *7* = sampling sites, *8* = path and M1–13 Morel et al. sites. (b) Schematic north–south cross-section in Abadla basin (redrawn from Doubinger and Fabre, 1983). *1* = Moscovian rhythmic terrigenous formation which is mainly marine, *2* = Westphalian C–D with coal facies, *3* = Lower unit of the redbeds formation probably stephanian between levels A and C and dated Autunian by flora fossils between levels E and H. *4* = Upper unit of the redbeds formation.


Abadla
Algeria
N
2°45
2°38
A
31°15
Garet El Hamra
Hadeb Chouahed
Oued
Guir
M 11
M 10
M 8, 9
Meksem El Khadem
7
M 12, 13
Koudiat
Zerigat
Garet El Betik
B
M 5, 7
M 1-4, 6
0
6 Km
Abadla
31°00
1 2 3 4 5 6 7 8

(a)


Chaib
Rassou
Garet el Hamra
Hadeb Chouahed
Koudiat Zerigat
A
B
1-6
12-13
8-11
7
800
700
A B C E F G H I
North
South
1
2
3
4
0 Km
5 Km

(b)

the components of the NRM. Selected thermal increments of 100ºC at lower temperatures and of 50–20º at higher ones were used in order to obtain a detailed picture of the NRM demagnetization. The low field magnetic susceptibility at room temperature K was measured after each heating step on a KLY-2 Kappabridge (Geofyzika, Brno). The results enable the detection of eventual magnetic mineralogy changes during stepwise thermal experiments. The magnetic carriers were studied by means of hysteresis cycles obtained with a translation inductometer associated with an electromagnet, able to induce field up to 1.6 T, and thermomagnetic curves obtained with a KLY-2 Kappabridge associated with a CS2 (Hrouda, 1994) apparatus (Geofyzika, Brno)

Demagnetization data for each specimen were plotted on orthogonal vector diagrams (Wilson and Everitt, 1963; Zijderveld, 1967) and on equal-area projections. Fisher statistics (Fisher, 1953) and bivariate distribution (Le Goff, 1990; Le Goff et al., 1992) statistics were used for data analysis.

4. Rock-magnetic properties and paleomagnetic measurements

The NRM intensities generally range from 0.75 to 14 $\times$ 10^{-2} A/m with a mean of 3.3 $\times$ 10^{-2} A/m. The magnetic susceptibility K at room temperature (Fig. 2) of most specimens underwent a weak decrease between the 300ºC and 550ºC heating steps. The magnetic susceptibility strongly increases at temperatures higher than 550ºC, probably due to formation of magnetite during heating. The rock-magnetism analysis pointed out clearly two different cases related to gray or green facies and much more numerous red facies.

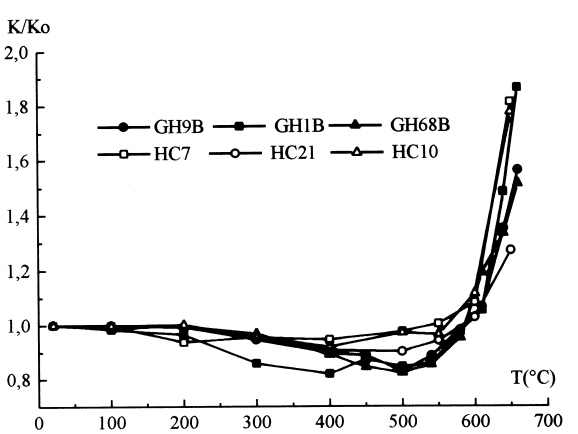


Fig. 2. Normalized magnetic susceptibility measured at room temperature after the thermal treatment.

4.1. Gray or green facies

The hysteresis loops (Fig. 3a) are mainly linear. The lines corresponding to increasing and decreasing fields are almost superimposed. There is therefore no visible saturation, even in high field, and it is likely

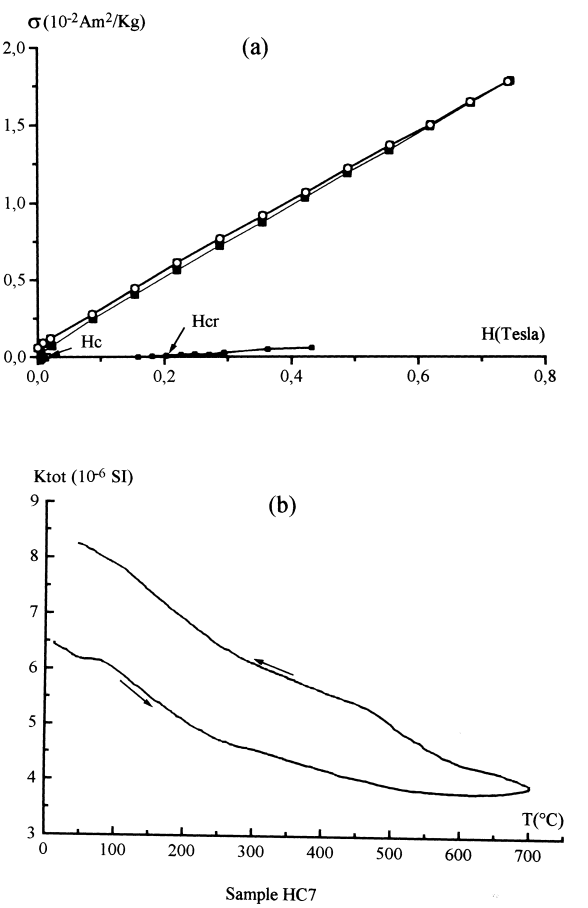


Fig. 3. Gray sample HC7. (a) Hysteresis curves of induced (full squares and empty circles). *Hc* = Coercive force; *Hcr* = remanent coercive force, σ = specific magnetization, *H* = induced field. (b) Thermomagnetic curves (heating and cooling have been done in the air).

that the paramagnetic minerals have a predominant effect in these weakly magnetic specimens. The high value of the remanent coercive force (0.2 T) and the lack of saturation suggest the presence of hematite or goethite. Thermomagnetic curves (Fig. 3b) confirm the dominance of the paramagnetic minerals by the hyperbolic nature of both heating and cooling curves. The slope variation on the cooling curve at 550–600ºC points out the formation of a new magnetic phase, which is probably magnetite. This facies accounts for nearly 15% of the studied specimens.

4.2. Red facies

The hysteresis loop curves (Fig. 4a) do not reach saturation here, but values during the decrease of the field exhibit stronger remanent magnetization than previously. The values of the remanent coercive force (Hcr) of specimens are high and range between 0.26 and 0.49 T. This indicates the existence of hematite or goethite as magnetic mineral. The presence of hematite is shown by the strong diminution of thermomagnetic curve for temperatures higher than 600ºC (Fig. 4b). Nedjari (1990) determined its percentage (7%) in the whole material. No evidence of the presence of goethite appears on the thermomagnetic curve, but other magnetic phases clearly also exist in some samples, perhaps from mineralogical changes during heating.

The 167 NRM vectors before any demagnetization treatment are scattered, around a great circle, between the direction of the present day magnetic field (PDMF) and a SSE sub-horizontal direction. This dispersion likely reflects the multivectorial nature of the remanence.

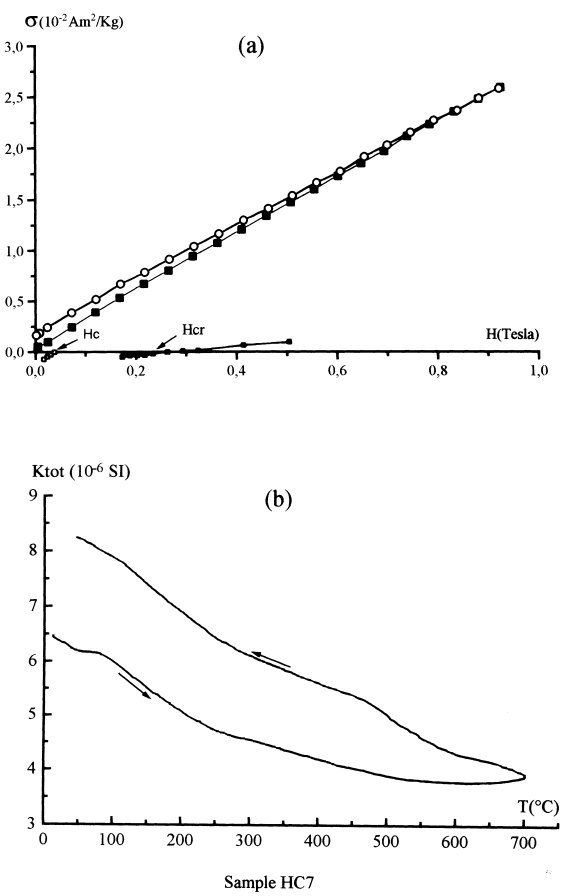


Fig. 4. Red sample HC16 (see legend of Fig. 3).

5. Paleomagnetic results

Thermal treatments confirmed the multivectorial nature of the NRM which consists of two components with unblocking temperature spectra completely separated for most specimens, partially or wholly overlapped in the others.

5.1. Components with separated unblocking temperature spectra

The most common magnetic type (57% of studied specimens) is characterized by separated unblocking temperature spectra of the two components. Orthogonal vectors diagrams (Fig. 5a) show a first component destroyed in most cases at temperature lower than 300ºC. The orientation of this component in geographic coordinates is consistent with the PDMF direction, so this component is likely of viscous nature (VRM). A characteristic remanent magnetization (ChRM) is identified at temperatures between 300 and 650–670ºC (red facies) or 300 and 550ºC (gray or green facies). Its direction is weakly inclined toward the SSE orientation.

The 96 ChRM directions are well grouped (Fig. 6 and Table 1) before ($D = 127.9^{o}$, $I = 17.4^{o}$, $k = 85$ and $\alpha_{95} = 1.5^{o}$) and after correction for bedding

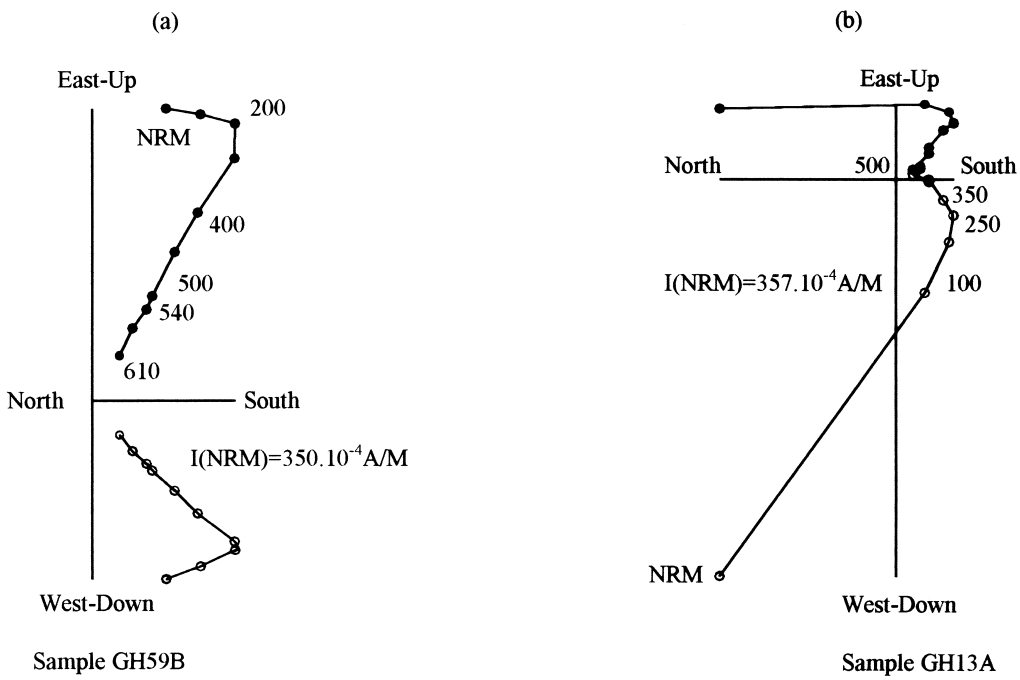


Fig. 5. Orthogonal vector plot (filled and open symbols correspond respectively to horizontal and vertical plane). (a) ChRM (sample GH59B). (b) Partially superimposed NRM components (sample GH13A).

($D = 129.6^o$, $I = 13.4^o$, $k = 93$ and $\alpha_{95} = 1.5^o$). The mean direction obtained by giving unit weight to each site is practically the same ($N = 11$ sites $D = 128.4^o$, $I = 17.4^o$, $k = 118$ and $\alpha_{95} = 3.9^o$ before tilting; $D = 130.1^o$, $I = 13.0^o$, $k = 138$ and $\alpha_{95} = 3.6^o$ after tilting). No fold test is available for these uniformly dipping beds. The application of the bivariate statistics (Le Goff, 1990; Le Goff et al., 1992) shows a circular distribution of the obtained 96 directions (ratio k_y/k_x equal to 1.2 and 1.0, respectively before and after dip correction). These ratio values indicate that the ChRM of the 96 specimens contain only a single component.

Samples of site 7 situated in the unfossiliferous beds (Meksem El Khadem) exhibited the same magnetic behavior. The mean magnetic direction of a well defined ChRM is for 18 specimens $D = 127.7^o$, $I = 8.1^o$, $k = 206$ and $\alpha_{95} = 2.3^o$ and $D = 128.5^o$, $I = 2.9^o$, $k = 206$ and $\alpha_{95} = 2.3^o$ respectively before and after dip correction (Table 1).

5.2. Partially superimposed magnetization components

For all the specimens of site 3, no ChRM has been obtained because the NRM results show two components with partially overlapping (Fig. 5b) unblocking temperature spectra (up to 650–670°C for the red facies). The corresponding great circles have a best intersection (McFadden and McElhinny, 1988) defined by $D = 133.9^o$, $I = 8.0^o$ and $\alpha_{95} = 5.1^o$ and $D = 134.3^o$, $I = 2.3^o$, $\alpha_{95} = 5.1^o$ respectively before and after dip correction (Table 1). In site 11, 12 great circles and 7 ChRMs have been determined according to the specimens. The ChRMs direction being in the plane of the great circles, the mean direction has been calculated combining these different data (McFadden and McElhinny, 1988): $D = 125.9^o$, $I = 8.7^o$ and $\alpha_{95} = 2.3^o$ and $D = 126.9^o$, $I = 6.0^o$ and $\alpha_{95} = 2.3^o$m respectively before and after dip correction (Table 1). The direction for these two sites is not significantly different from the direction at the other sites. The low blocking temperature component could be a chemical magnetic overprint of relatively recent age.

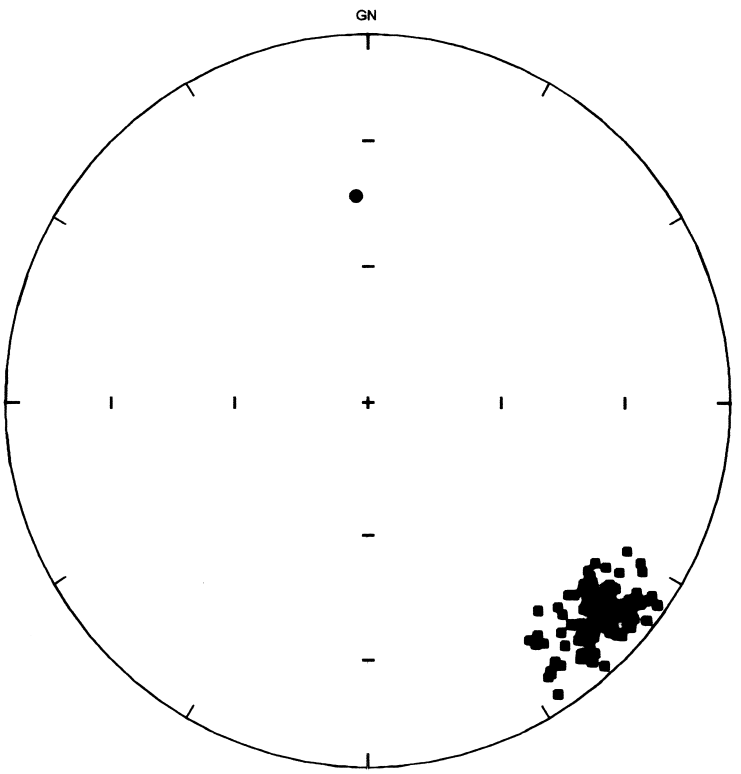


Fig. 6. Equal area projection plot (filled symbols correspond to lower hemisphere). Magnetic mean directions of 96 Autunian specimens after dip correction. Filled circle indicates the present Earth magnetic field direction.

5.3. Totally superimposed magnetization components

For 29 red specimens, mainly from sites 12 and 13, the mean magnetic direction of the NRM after analysis are displayed along a partial girdle on an equal area projection. They are in an intermediate position between the PDMF and a SSE direction. This scattering shows that their NRM included at least two completely superimposed components with exactly superimposed unblocking temperature spectra (until 650–670ºC for the red facies). These directions have been eliminated from the directional statistic.

5.4. Age of the magnetization

An important remark can be made about the magnetic mean directions obtained at different levels of the two units of the Abadla formation. In the lower unit (Autunian dated beds), paleomagnetic samples have been collected from the lower levels (Garet El Hamra, GEH, sites 1–6 and 12–13) and the upper levels (Hadeb Chouahed, HC, sites 8–11). In the upper unit (undated beds), samples have been taken from the middle levels (Meksem El Khadem, MEK, site 7 and sites 8–13 of Morel et al. (1981)) and from the upper levels (South Garet El Batik, GEB, sites 1–7 of Morel et al. (1981)). The obtained directions of these four levels are different before or after dip correction and their values after dip correction are as follows:

GEH	7 sites	$D = 130.6^{o}$	$I = 15.4^{o}$	$\alpha_{95} = 4.4^{o}$
HC	4 sites	$D = 129.1^{o}$	$I = 9.0^{o}$	$\alpha_{95} = 1.9^{o}$
MEK	7 sites	$D = 131.0^{o}$	$I = 5.7^{o}$	$\alpha_{95} = 5.3^{o}$
GEB	7 sites	$D = 126.5^{o}$	$I = 14.9^{o}$	$\alpha_{95} = 7.1^{o}$

Except for levels of GEB, the variation of the magnetic direction from the bottom to the top of the formation is significant (no overlapping α_{95}) between bottom (GEH) and top (MEK) of the succession. The

Table 1
Number of direction (N), ChRM mean direction (D, I) and virtual geomagnetic pole (Lt, Lg)

Before dip correction						After dip correction				Pole coordinates			
Site	N	D (°)	I (°)	k	α_{95} (°)	D (°)	I (°)	k	α_{95} (°)	Lt (°S)	Lg (°E)	K	A_{95} (°)
Lower unit of the Abadla formation (Autunian dated)													
ChRM													
1	8	127.1	23.4	107	4.8	128.9	18.2	107	4.8	26.5	56.4		
2	4	127.2	27.4	198	5.0	129.4	22.2	198	5.0	25.3	54.2		
5	4	125.0	25.0	106	6.8	128.1	19.0	106	6.8	25.6	56.6		
6	8	129.3	20.0	90	5.2	131.5	13.1	90	5.2	30.2	56.7		
8	12	124.5	12.1	187	3.0	125.8	8.3	187	3.0	27.5	63.0		
9	10	135.3	12.8	191	3.2	136.5	9.4	189	3.2	35.1	54.3		
10	12	126.4	14.0	226	2.7	128.0	11.2	226	2.7	28.2	60.2		
12	15	129.2	21.6	118	3.3	131.7	17.8	118	3.3	28.7	54.4		
13	16	127.7	18.3	123	3.2	129.9	14.8	123	3.2	28.4	57.2		
7 ChRM + 12 great circles													
11	19	125.9	8.7		2.3	126.9	6.0		2.3	29.1	63.3		
Great circles													
3	8	133.9	8.0		5.1	134.3	2.3		5.1	35.9	59.3		
Mean direction of sites (ChRM + great circles)													
	11	128.4	17.4	118	3.9	130.1	13.0	138	3.6	29.2	57.8		
	11									29.1	57.8	462	2.0
Mean direction of specimens (ChRM)													
	96	127.9	17.4	85	1.5	129.6	13.4	93	1.5	28.8	57.8		
										30.8	55.0	88	1.5
Upper unit of the Abadla formation (undated)													
7	18	127.7	8.1	206	2.3	128.5	2.9	206	2.3	31.3	63.6		
Combined data (site 7 + sites 1–13 of Morel et al., 1981)													
	14	127.0	16.5	60	4.8	128.8	10.3	60	4.8	29.1	60.0		
										29.2	60.0	81	4.2

k, K, α_{95} and A_{95} are the Fisher's parameters (Fisher, 1953).

magnetization was not acquired at the same period, at least in these two parts of the succession, and should not be therefore a remagnetization. Moreover, the paleomagnetic direction obtained in the Koudiat Zerigat (Kies, 1994) is very different from our result, and that also excludes the possibility of a magnetic overprint related to this volcanic event.

6. Paleopoles

The mean paleopole of the Autunian dated unit, determined by giving unit weight to each of the eleven sites (ChRM and great circles), is situated at 29.1°S, 57.8°E with $K = 462$ and $A_{95} = 2.0°$ (Table 1). This datum is not significantly different from that determined by giving unit weight to each of the obtained ChRM mean directions ($N = 96$ ChRM, 28.8°S, 57.8°E; Table 1). Finally, site 7 combined with data of Morel et al. (1981) gave a paleopole situated at 29.2°S, 60.0°E (Table 1), close to our new Autunian pole. The upper unit of the Abadla formation is therefore of Permian (likely Autunian) age.

Table 2 summarizes the reliable paleopoles from the Moscovian to Permian for Algerian Sahara craton. It appears that the new Autunian paleopole of Abadla is consistent with the Stephano-Autunian poles of Tiguentourine and of El Adeb Larache. We can point out the intermediate position of the paleopole of this study between those of the Moscovian pole and the Permian overprints, which could be of Autunian to upper Permian age.

Table 2
Moscovian to Permian paleomagnetic pole positions for stable Saharan craton

Formation	Age	Lt (ºS)	Lg (ºE)	A_{95} (º)	Reference
El Adeb Larache	Moscovian	28.7	55.8	2.9	Henry et al. (1992)
El Adeb Larache	Stephano-Autunian	38.5	57.5	2.1	Henry et al. (1992)
Lower Tiguentourine	Stephano-Autunian	33.8	61.4	4.0	Derder et al. (1994)
Abadla (Lower unit)	Autunian	29.1	57.8	2.0	This study
Abadla (Upper unit)	?	29.0	60.0	5.0	Morel et al. (1981)
Hassi Bachir	Permian overprint	35.5	60.0	4.6	Daly and Irving (1983)
El Adeb Larache	Permian overprint	42.0	65.1	2.6	Henry et al. (1992)

7. Dynamical implications

This result has interesting dynamical implications concerning the subsidence and accumulation rate of the sediments in the Abadla basin. This basin has been formed during a period of orogenic calm and strong subsidence consecutive to an earlier Hercynian phase. As it is open to the west, the 2000-m thick succession represents a minimum estimate of the total thickness of sediments deposited. The paleomagnetic result indicates that the subsidence and simultaneous sedimentary filling would have occurred during the Autunian, lasting no more than 20–25 Myr.

8. Conclusion

The Autunian paleopole (Lat = 29.1ºS, Long = 57.8ºE, A_{95} = 2.0º) obtained in these levels represents a new contribution for the precise definition of the Autunian pole of the African apparent polar wander path. This new datum is close to that of the upper unit of the redbeds formation. This narrow proximity of the two paleopoles enables us to reduce the age uncertainty window of the unfossiliferous upper part. This part has thus probably an Autunian age, giving for the subsidence and filling of the Abadla basin a duration no more than 20–25 Myr.

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